AI Encyclopedia-Nutritional Analysis

1. Function Description

After the program starts, ask the large model which fruit on the desktop has the highest vitamin content. After thinking, the large model will reply with the answer to the question, and the program will control the robotic arm to point to that fruit.

2. Startup

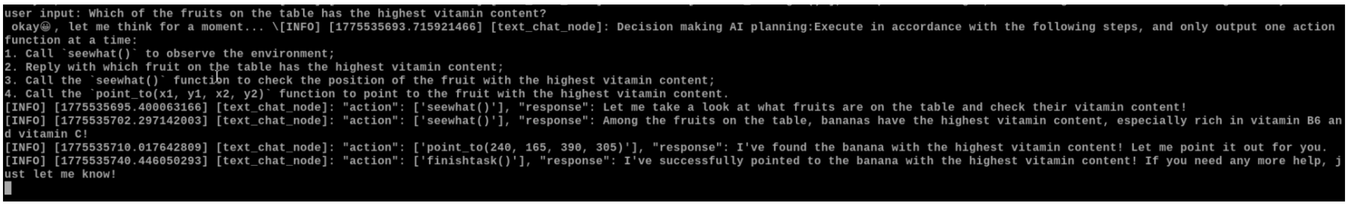
After entering the Docker container, enter the following commands in the Docker terminal.

```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and input the following command:

```
ros2 run text_chat text_chat
```

Then input "Which of the fruits on the table has the highest vitamin content?" in the text_chat terminal and press Enter; if it's the voice version, wake up the voice module and directly say to the voice module "Which of the fruits on the table has the highest vitamin content?", then wait for the large model to think and reply, as shown in the figure below:



```
user input: Which of the fruits on the table has the highest vitamin content?
 okay😊, let me think for a moment... \[INFO] [1775535693.715921466] [text_chat_node]: Decision making AI planning:Execute in accordance with the following steps, and only output one action function at a time:
1. Call `seewhat()` to observe the environment;
2. Reply with which fruit on the table has the highest vitamin content;
3. Call the `seewhat()` function to check the position of the fruit with the highest vitamin content;
4. Call the `point_to(x1, y1, x2, y2)` function to point to the fruit with the highest vitamin content.
[INFO] [1775535695.400063166] [text_chat_node]: "action": ['seewhat()'], "response": Let me take a look at what fruits are on the table and check their vitamin content!
[INFO] [1775535702.297142003] [text_chat_node]: "action": ['seewhat()'], "response": Among the fruits on the table, bananas have the highest vitamin content, especially rich in vitamin B6 and vitamin C!
[INFO] [1775535710.017642809] [text_chat_node]: "action": ['point_to(240, 165, 390, 305)'], "response": I've found the banana with the highest vitamin content! Let me point it out for you.
[INFO] [1775535740.446050293] [text_chat_node]: "action": ['finishtask()'], "response": I've successfully pointed to the banana with the highest vitamin content! If you need any more help, just let me know!
```

Then the robotic arm will point to the banana:

3. Task Planning

1. Call `seewhat()` to observe the environment;
2. Reply with which fruit on the table has the highest vitamin content;
3. Call the `seewhat()` function to check the position of the fruit with the highest vitamin content;
4. Call the `point_to(x1, y1, x2, y2)` function to point to the fruit with the highest vitamin content.